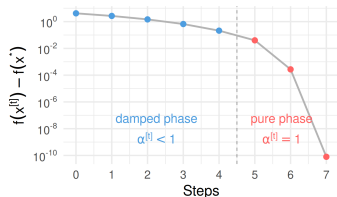
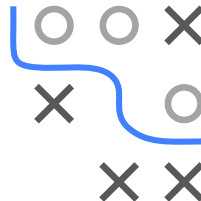


Optimization in Machine Learning

Second order methods

Deep Dive: Convergence of Newton-Raphson



Learning goals

- Assumptions
- Damped phase
- Pure Newton phase
- Complexity bound

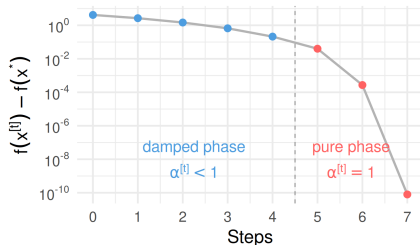
Slides based on

► Boyd and Vandenberghe 2004 (Ch. 9.5.3)

SETTING

- Recall:

- with full steps ($\alpha^{[t]} = 1$), NR may overshoot/diverge far from $\mathbf{x}^* \Rightarrow$ damping is necessary
- many objective functions are only locally quadratic
- In this deep dive, we show that damped NR has two phases:
 - **Damped phase:** quadratic model unreliable, $\alpha^{[t]} < 1$ possible; f decreases by a *constant* δ per step
 - **Pure Newton phase:** quadratic model accurate, $\alpha^{[t]} = 1$ always accepted; *quadratic* convergence

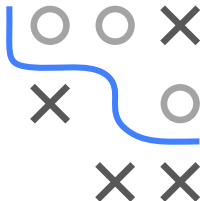


Example 2 (cont'd):

$$f(\mathbf{x}) = (1 + x_1^2)^{1/2} + \frac{x_2^2}{2}$$

from $\mathbf{x}^{[0]} = (3, 2)^\top$,

with $\gamma_1 = 0.1$, $\tau = 0.7$



SETTING

- **Assumptions** on the sublevel set $\mathcal{S} = \{\mathbf{x} : f(\mathbf{x}) \leq f(\mathbf{x}^{[0]})\}$:
 - $f \in \mathcal{C}^2$, m -strongly convex ($\mathbf{H}(\mathbf{x}) \succeq m\mathbf{I}$) and L -smooth ($\mathbf{H}(\mathbf{x}) \preceq L\mathbf{I}$)
 - $\mathbf{H}$ is **Lipschitz** with constant $L_{\mathbf{H}}$ ($\rightarrow$ bound on the 3rd derivative):

$$\|\mathbf{H}(\mathbf{x}) - \mathbf{H}(\tilde{\mathbf{x}})\|_2 \leq L_{\mathbf{H}}\|\mathbf{x} - \tilde{\mathbf{x}}\| \quad \text{for all } \mathbf{x}, \tilde{\mathbf{x}} \in \mathcal{S}$$

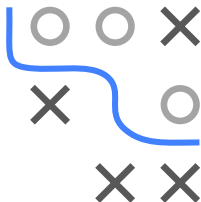
Intuition: $L_{\mathbf{H}}$ measures how well the quadratic model approximates f ($L_{\mathbf{H}} = 0$ for quadratic f)

- Damped NR with backtracking: $\alpha_{\text{init}} = 1$, $\gamma_1 \in (0, 1/2)$, $\tau \in (0, 1)$

Theorem (Convergence of damped NR). With the scaled gradient norm $\theta^{[t]} = \frac{L_{\mathbf{H}}}{2m^2} \|\nabla f(\mathbf{x}^{[t]})\|$, there exist $\eta \in (0, m^2/L_{\mathbf{H}}]$ and $\delta > 0$ such that

$$\|\nabla f(\mathbf{x}^{[t]})\| \geq \eta \Rightarrow f(\mathbf{x}^{[t+1]}) - f(\mathbf{x}^{[t]}) \leq -\delta \quad (\text{damped})$$

$$\|\nabla f(\mathbf{x}^{[t]})\| < \eta \Rightarrow \alpha^{[t]} = 1 \text{ and } \theta^{[t+1]} \leq (\theta^{[t]})^2 \quad (\text{pure})$$



WHAT THE THEOREM IMPLIES: PURE PHASE

Before we actually prove the theorem, we analyze its implications for the convergence of damped NR.

- **Message 1:** once NR enters the pure phase, it stays there

- suppose we are in the pure phase at step t , i.e. $\|\nabla f(\mathbf{x}^{[t]})\| < \eta$
- since $\eta \leq m^2/L_H$ by def., the scaled gradient norm is below $\frac{1}{2}$:

$$\theta^{[t]} = \frac{L_H}{2m^2} \|\nabla f(\mathbf{x}^{[t]})\| < \frac{L_H}{2m^2} \cdot \frac{m^2}{L_H} = \frac{1}{2}$$

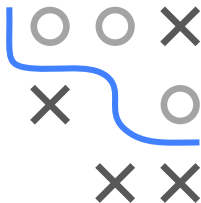
- with this and the theorem:

$$\theta^{[t+1]} \leq (\theta^{[t]})^2 = \theta^{[t]} \cdot \theta^{[t]} < \frac{1}{2} \theta^{[t]}$$

$$\Rightarrow \|\nabla f(\mathbf{x}^{[t+1]})\| < \frac{1}{2} \|\nabla f(\mathbf{x}^{[t]})\| < \eta$$

- i.e., the condition is also satisfied at step $t + 1$ and by recursion at all future steps $l \geq t$, implying (by the theorem):

$$\alpha^{[l]} = 1 \quad \text{and} \quad \theta^{[l+1]} \leq (\theta^{[l]})^2$$



WHAT THE THEOREM IMPLIES: PURE PHASE

- **Message 2:** convergence is extremely rapid in the pure phase

- $\theta^{[t+1]} \leq (\theta^{[t]})^2$ is exactly **quadratic convergence**
- Applying it recursively, for $l \geq t$:

$$\theta^{[l]} \leq (\theta^{[t]})^{2^{l-t}} \leq \left(\frac{1}{2}\right)^{2^{l-t}}$$

- Strong convexity gives $f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{1}{2m} \|\nabla f(\mathbf{x})\|^2$ and thus:

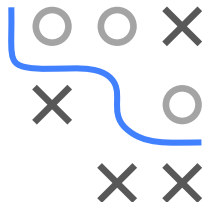
$$f(\mathbf{x}^{[l]}) - f(\mathbf{x}^*) \leq \epsilon_0 (\theta^{[l]})^2 \leq \epsilon_0 \left(\frac{1}{2}\right)^{2^{l-t+1}} \quad \epsilon_0 = \frac{2m^3}{L_H^2}$$

- Inverting this for the step count yields:¹

$$f(\mathbf{x}^{[l]}) - f(\mathbf{x}^*) \leq \epsilon \quad \text{after at most} \quad \log_2 \log_2(\epsilon_0/\epsilon) \quad \text{pure steps}$$

⇒ **Note:** at most 6 pure steps give an accuracy of $\epsilon \approx 5 \cdot 10^{-20} \epsilon_0$

¹Exactly $\log_2 \log_2(\epsilon_0/\epsilon) - 1$ steps; the -1 is dropped for readability.



WHAT THE THEOREM IMPLIES: DAMPED PHASE & FINAL RESULT

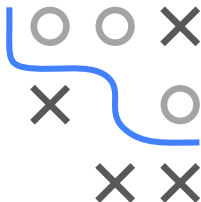
- **Message 3:** the number of steps in the damped phase is bounded
 - Every damped step reduces f by at least δ , so after T damped steps $f(\mathbf{x}^{[T]}) \leq f(\mathbf{x}^{[0]}) - T\delta$
 - Since f cannot decrease below $f(\mathbf{x}^*)$:

$$\#\{\text{damped steps}\} \leq \frac{f(\mathbf{x}^{[0]}) - f(\mathbf{x}^*)}{\delta}$$

$\Rightarrow$ finite, but expensive phase; grows with the initial gap

- **Final result** (both phases together): to reach $f(\mathbf{x}^{[t]}) - f(\mathbf{x}^*) \leq \epsilon$:

$$\#\text{iterations} \leq \underbrace{\frac{f(\mathbf{x}^{[0]}) - f(\mathbf{x}^*)}{\delta}}_{\text{damped}} + \underbrace{\log_2 \log_2(\epsilon_0/\epsilon)}_{\text{pure, } \approx 6}$$



PROOF STRATEGY

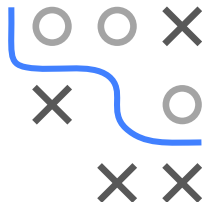
Note: we have now seen the implications of the theorem, but have not yet proven it, i.p. the existence of η and δ . Our proof strategy is as follows:

- ❶ **Tool:** the **Newton decrement** $\lambda(\mathbf{x})$ as progress measure
- ❷ **Damped phase:** L -smoothness bounds the step size from below

$$\alpha^{[t]} \geq \tau \frac{m}{L} \implies \delta = \gamma_1 \tau \eta^2 \frac{m}{L^2}$$

- ❸ **Pure phase:** Lipschitz $\mathbf{H}$ keeps the quadratic model accurate enough that

$$\alpha^{[t]} = 1 \text{ is accepted} \quad \text{and} \quad \theta^{[t+1]} \leq (\theta^{[t]})^2$$
$$\text{for } \eta = \min\{1, 3(1 - 2\gamma_1)\} \frac{m^2}{L_{\mathbf{H}}}$$



TOOL: THE NEWTON DECREMENT

- Backtracking needs $\nabla f(\mathbf{x})^\top \mathbf{d}$ along $\mathbf{d} = -\mathbf{H}(\mathbf{x})^{-1} \nabla f(\mathbf{x})$; its negative is the squared **Newton decrement** $\lambda(\mathbf{x})$:

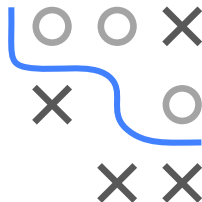
$$\lambda(\mathbf{x})^2 = \underbrace{\nabla f(\mathbf{x})^\top \mathbf{H}(\mathbf{x})^{-1} \nabla f(\mathbf{x})}_{\text{definition}} = \underbrace{\mathbf{d}^\top \mathbf{H}(\mathbf{x}) \mathbf{d}}_{\text{squared } \mathbf{H}\text{-norm of } \mathbf{d}} = \underbrace{-\nabla f(\mathbf{x})^\top \mathbf{d}}_{\text{-- dir. derivative}}$$

$\Rightarrow$ the Armijo rule for the Newton step reads $f(\mathbf{x} + \alpha \mathbf{d}) \leq f(\mathbf{x}) - \gamma_1 \alpha \lambda(\mathbf{x})^2$

$\Rightarrow$ λ measures the gradient in the geometry of $\mathbf{H}$, not the Euclidean one

- From $m\mathbf{I} \preceq \mathbf{H}(\mathbf{x}) \preceq L\mathbf{I}$, i.e. $\frac{1}{L}\mathbf{I} \preceq \mathbf{H}(\mathbf{x})^{-1} \preceq \frac{1}{m}\mathbf{I}$:

$$\frac{1}{L} \|\nabla f(\mathbf{x})\|^2 \leq \lambda(\mathbf{x})^2 \leq \frac{1}{m} \|\nabla f(\mathbf{x})\|^2 \quad \|\mathbf{d}\| \leq \frac{\lambda(\mathbf{x})}{\sqrt{m}}$$



DAMPED PHASE: THE STEP SIZE CANNOT GET SMALL

- Assume $\|\nabla f(\mathbf{x}^{[t]})\| \geq \eta$ and abbreviate $\mathbf{d} = \mathbf{d}^{[t]}$, $\lambda = \lambda(\mathbf{x}^{[t]})$
- **Recall** the descent lemma (L -smoothness) along $\mathbf{d}$, for any $\alpha \geq 0$:

$$f(\mathbf{x}^{[t]} + \alpha \mathbf{d}) \leq f(\mathbf{x}^{[t]}) + \alpha \nabla f(\mathbf{x}^{[t]})^\top \mathbf{d} + \frac{L}{2} \alpha^2 \|\mathbf{d}\|^2 \leq f(\mathbf{x}^{[t]}) - \alpha \lambda^2 + \frac{L}{2m} \alpha^2 \lambda^2$$

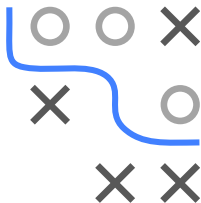
using $\nabla f(\mathbf{x}^{[t]})^\top \mathbf{d} = -\lambda^2$ and $\|\mathbf{d}\|^2 \leq \lambda^2/m$

- $\hat{\alpha} = m/L$ minimizes the right side, and because $\gamma_1 < \frac{1}{2}$:

$$f(\mathbf{x}^{[t]} + \hat{\alpha} \mathbf{d}) \leq f(\mathbf{x}^{[t]}) - \frac{m}{2L} \lambda^2 \leq f(\mathbf{x}^{[t]}) - \gamma_1 \hat{\alpha} \lambda^2$$

$\Rightarrow \hat{\alpha}$ passes the Armijo rule, and so does every smaller step; backtracking therefore accepts before shrinking past $\tau \hat{\alpha}$:

$$\alpha^{[t]} \geq \tau \frac{m}{L}$$



DAMPED PHASE: GUARANTEED DECREASE

- The accepted step satisfies the Armijo rule, hence

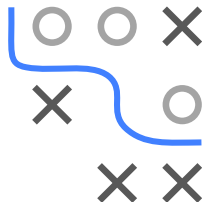
$$\begin{aligned}f(\mathbf{x}^{[t+1]}) - f(\mathbf{x}^{[t]}) &\leq -\gamma_1 \alpha^{[t]} \lambda^2 \\&\leq -\gamma_1 \tau \frac{m}{L} \lambda^2 \\&\leq -\gamma_1 \tau \frac{m}{L^2} \|\nabla f(\mathbf{x}^{[t]})\|^2 \\&\leq -\gamma_1 \tau \frac{m}{L^2} \eta^2\end{aligned}$$

- Arguments:** Armijo rule (line 1). Step size bound (line 1 $\rightarrow$ 2).

$\lambda^2 \geq \|\nabla f\|^2 / L$ (line 2 $\rightarrow$ 3). Phase condition (line 3 $\rightarrow$ 4).

$$\delta = \gamma_1 \tau \eta^2 \frac{m}{L^2}$$

$\Rightarrow$ the decrease does not shrink as t grows, making Message 3 possible



PURE PHASE: A CUBIC BOUND ALONG THE NEWTON RAY

- Now assume $\|\nabla f(\mathbf{x}^{[t]})\| < \eta$ and restrict f to the Newton ray:
 $\tilde{f}(\alpha) = f(\mathbf{x}^{[t]} + \alpha \mathbf{d})$, so $\tilde{f}'(0) = -\lambda^2$ and $\tilde{f}''(\alpha) = \mathbf{d}^\top \mathbf{H}(\mathbf{x}^{[t]} + \alpha \mathbf{d}) \mathbf{d}$
- By the Lipschitz condition between $\mathbf{x}^{[t]} + \alpha \mathbf{d}$ and $\mathbf{x}^{[t]}$, then sandwiched with $\mathbf{d}$ (Cauchy-Schwarz):

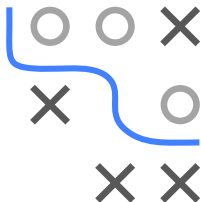
$$\begin{aligned}\|\mathbf{H}(\mathbf{x}^{[t]} + \alpha \mathbf{d}) - \mathbf{H}(\mathbf{x}^{[t]})\|_2 &\leq \alpha L_{\mathbf{H}} \|\mathbf{d}\| \\ |\mathbf{d}^\top (\mathbf{H}(\mathbf{x}^{[t]} + \alpha \mathbf{d}) - \mathbf{H}(\mathbf{x}^{[t]})) \mathbf{d}| &\leq \alpha L_{\mathbf{H}} \|\mathbf{d}\|^3\end{aligned}$$

- This is $|\tilde{f}''(\alpha) - \tilde{f}''(0)|$; with $\tilde{f}''(0) = \lambda^2$, $\|\mathbf{d}\| \leq \lambda/\sqrt{m}$:

$$\tilde{f}''(\alpha) \leq \lambda^2 + \alpha \frac{L_{\mathbf{H}}}{m^{3/2}} \lambda^3$$

- Integrating twice (with $\tilde{f}'(0) = -\lambda^2$) and setting $\alpha = 1$:

$$f(\mathbf{x}^{[t]} + \mathbf{d}) \leq f(\mathbf{x}^{[t]}) - \frac{1}{2} \lambda^2 + \frac{L_{\mathbf{H}}}{6m^{3/2}} \lambda^3$$



PURE PHASE: THE FULL STEP IS ACCEPTED

- Strong convexity turns the phase condition into a bound on λ : if

$$\eta \leq 3(1 - 2\gamma_1) \frac{m^2}{L_H}, \text{ then}$$

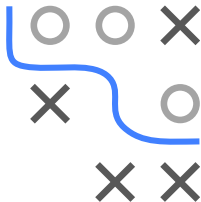
$$\lambda \leq \frac{\|\nabla f(\mathbf{x}^{[t]})\|}{\sqrt{m}} < \frac{\eta}{\sqrt{m}} \leq 3(1 - 2\gamma_1) \frac{m^{3/2}}{L_H} \implies \frac{L_H \lambda}{6m^{3/2}} \leq \frac{1 - 2\gamma_1}{2}$$

- Factoring the cubic bound, the bracket is therefore at least γ_1 :

$$\begin{aligned} f(\mathbf{x}^{[t]} + \mathbf{d}) &\leq f(\mathbf{x}^{[t]}) - \lambda^2 \underbrace{\left(\frac{1}{2} - \frac{L_H \lambda}{6m^{3/2}} \right)}_{\geq \gamma_1} \leq f(\mathbf{x}^{[t]}) - \gamma_1 \lambda^2 \\ &= f(\mathbf{x}^{[t]}) + \gamma_1 \cdot 1 \cdot \nabla f(\mathbf{x}^{[t]})^\top \mathbf{d} \end{aligned}$$

$\Rightarrow$ this is exactly the Armijo rule at $\alpha = 1$: **backtracking accepts the full step straight away**

$\Rightarrow$ here $\gamma_1 < \frac{1}{2}$ is essential: otherwise $3(1 - 2\gamma_1) \leq 0$ and no such η exists



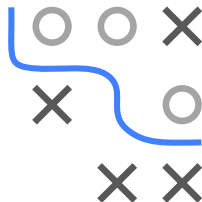
PURE PHASE: QUADRATIC CONVERGENCE

- The Newton step solves $\nabla f(\mathbf{x}^{[t]}) + \mathbf{H}(\mathbf{x}^{[t]})\mathbf{d} = \mathbf{0}$, so we can subtract it:

$$\begin{aligned}\|\nabla f(\mathbf{x}^{[t+1]})\| &= \|\nabla f(\mathbf{x}^{[t]} + \mathbf{d}) - \nabla f(\mathbf{x}^{[t]}) - \mathbf{H}(\mathbf{x}^{[t]})\mathbf{d}\| \\ &= \left\| \int_0^1 (\mathbf{H}(\mathbf{x}^{[t]} + \alpha\mathbf{d}) - \mathbf{H}(\mathbf{x}^{[t]}))\mathbf{d} \, d\alpha \right\| \leq \frac{L_{\mathbf{H}}}{2} \|\mathbf{d}\|^2 \\ &= \frac{L_{\mathbf{H}}}{2} \|\mathbf{H}(\mathbf{x}^{[t]})^{-1} \nabla f(\mathbf{x}^{[t]})\|^2 \leq \frac{L_{\mathbf{H}}}{2m^2} \|\nabla f(\mathbf{x}^{[t]})\|^2\end{aligned}$$

- **Arguments:** fundamental theorem of calculus (line 1 $\rightarrow$ 2). Lipschitz $\mathbf{H}$ and $\int_0^1 \alpha \, d\alpha = \frac{1}{2}$ (line 2). $\|\mathbf{H}^{-1}\| \leq 1/m$ (line 3).
- Multiplying by $\frac{L_{\mathbf{H}}}{2m^2}$ gives the theorem's form; taking the smaller of the two ceilings on η (previous slide and theorem/Message 1):

$$\theta^{[t+1]} \leq (\theta^{[t]})^2 \quad \eta = \min\{1, 3(1 - 2\gamma_1)\} \frac{m^2}{L_{\mathbf{H}}}$$



PUTTING EVERYTHING TOGETHER

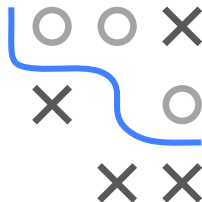
- Substituting η into $\delta = \gamma_1 \tau \eta^2 \frac{m}{L^2}$:

$$\delta = \gamma_1 \tau \min\{1, 9(1 - 2\gamma_1)^2\} \frac{m^5}{L^2 L_H^2}$$

- Inserting this into the final result and using $\log_2 \log_2(\epsilon_0/\epsilon) \approx 6$:

$$\text{\#iterations} \leq 6 + \frac{L^2 L_H^2 / m^5}{\gamma_1 \tau \min\{1, 9(1 - 2\gamma_1)^2\}} (f(\mathbf{x}^{[0]}) - f(\mathbf{x}^*))$$

- the accuracy ϵ has collapsed into the constant 6
- the cost is driven by how far the starting point is from the optimum
- all the cost sits in the damped phase, with $\frac{L^2 L_H^2}{m^5} = \kappa^2 (L_H / m^{3/2})^2$:
ill-conditioning ($\kappa = L/m$) and how far f is from quadratic
- But:** m, L, L_H are unknown for any real problem $\Rightarrow$ a purely theoretical result, never evaluated in practice



DISCUSSION

- Slideset 1 claimed NR converges *locally* quadratically
- “**Locally**” is the pure phase, and we quantified the neighborhood:

$$\|\nabla f(\mathbf{x}^{[t]})\| < \eta = \min\{1, 3(1 - 2\gamma_1)\} \frac{m^2}{L_H}$$

- “**Quadratically**”: we showed that $\theta^{[t+1]} \leq (\theta^{[t]})^2$, i.e., the scaled gradient norm decreases quadratically
- **Damped NR**: converges from any $\mathbf{x}^{[0]}$ whose sublevel set satisfies the assumptions, but potentially slowly until the pure phase is reached
- **Undamped NR** ($\alpha^{[t]} = 1$ always): the quadratic rate never used backtracking, only that a full step was taken $\Rightarrow$ quadratic convergence still holds for undamped NR inside the local neighborhood; but: the damped-phase argument collapses, nothing forces f to decrease outside the neighborhood

